

D3 3. 16-20

12	15	14	14	14	18	17	15	17	19
13	17	19	16	15	18	20	20	22	13
13	17	12	16	15	18	20	17	12	13
12	17	19	16	15	18	20	19	12	13

x_i	12	13	14	15	16	17	18	19	20
h_i	6	5	3	5	3	5	4	5	4

$$h = 40$$

$$1) \bar{x}_h = \frac{1}{n} \sum x_i h_i = \frac{1}{40} (12 \cdot 6 + 13 \cdot 5 + 14 \cdot 3 + 15 \cdot 5 + 16 \cdot 3 + 17 \cdot 5 + 18 \cdot 4 + 19 \cdot 5 + 20 \cdot 4) = \frac{634}{40} \approx 15,85$$

$$2) \frac{1}{40} \sum x_i^2 h_i = (12^2 \cdot 6 + 13^2 \cdot 5 + 14^2 \cdot 3 + 15^2 \cdot 5 + 16^2 \cdot 3 + 17^2 \cdot 5 + 18^2 \cdot 4 + 19^2 \cdot 5 + 20^2 \cdot 4) = \frac{10336}{40} \approx 258,4$$

$$\Delta_f = 258,4 - (15,85)^2 = 258,4 - 251,22 \approx 4,18$$

$$S^2 = \frac{n}{n-1} D_0 = \frac{40}{39} 7,18 \approx 7,36$$

$$S = \sqrt{7,36} \approx 2,71$$

$$7) m = 2 + 3,322 \lg |n| \approx 2 + 3,322 \lg 40 \approx 6,32 \approx 6$$

$$h = \frac{x_{\max} - x_{\min}}{m} \approx \frac{20 - 12}{6} \approx 1,33$$

Для зручності обчислимо: $m = 5$ $h = 2$

Інтер.	$[12; 14)$	$[14; 16)$	$[16; 18)$	$[18; 20)$	$[20; 22)$
n_i	11	8	8	9	4

$$2) \sigma = 2 \quad \gamma = 0,95 : \left(\bar{x}_0 - t \frac{\sigma}{\sqrt{n}}; \bar{x}_0 + t \frac{\sigma}{\sqrt{n}} \right)$$

Функц. Лапласа; $\Phi(t) = \gamma/2 = 0,475 \Rightarrow t = 1,96$

$$t \frac{\sigma}{\sqrt{n}} = 1,96 \cdot \frac{2}{\sqrt{40}} \approx 1,96 \cdot \frac{2}{\sqrt{6,32}} \approx 0,62$$

$$(15,85 - 0,62) = 15,23$$

$$(15,85 + 0,62) = 16,47$$

$$(15,23; 16,47)$$

$$3) \gamma = 0,99 \quad \left(\bar{x}_0 - t_{\gamma} \frac{s}{\sqrt{n}} ; \bar{x}_0 + t_{\gamma} \frac{s}{\sqrt{n}} \right)$$

$$k = n - 1 = 39, \text{ За табл. } t_{\gamma} \approx 2,71$$

$$t_{\gamma} \frac{s}{\sqrt{n}} = 2,71 \cdot \frac{2,71}{6,32} \approx 1,16$$

$$75,85 + 1,16 = 77,01$$

$$75,85 - 1,16 = 74,69$$

$$(74,69 ; 77,01)$$

$$n) F(x_1^2) = \frac{7 - 0,95}{2} = 0,025 \Rightarrow x_1^2 \approx 58,1$$

$$F(x_2^2) = \frac{7 + 0,95}{2} = 0,975 \Rightarrow x_2^2 \approx 23,7$$

$$(n-1)s^2 = 39 \cdot 7,36 = 287,04$$

$$287,04 / 58,1 \approx 4,94$$

$$287,04 / 23,7 \approx 12,11$$

$$(4,94 ; 12,11)$$

$$\sqrt{4,94} \approx 2,22$$

$$\sqrt{12,11} \approx 3,48$$

$$(2,22 ; 3,48)$$